

6. Levels of bacteria in milk are routinely monitored.

(a) Most milk sold to consumers is pasteurised and homogenised.

(i) Explain why milk is pasteurised. [2]

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(ii) Describe the process of homogenisation and explain why it is carried out. [4]

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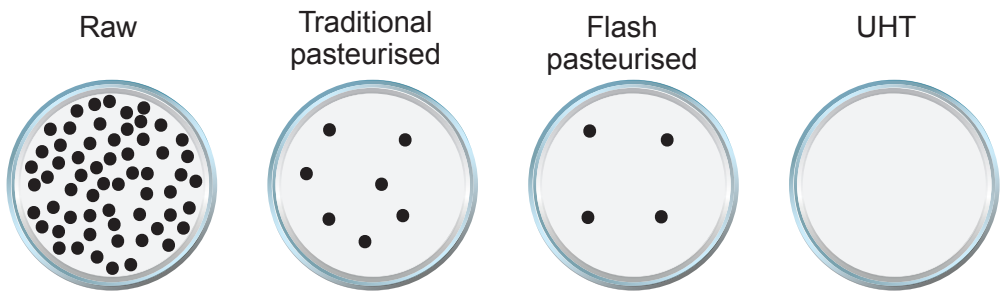
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(b) Scientists investigated four types of milk. They monitored the growth of bacterial colonies on agar plates using 0.1 cm³ from each type of milk over two days. Their results are shown below.



Type of milk	Heat treatment	Number of bacterial colonies in a 40 cm ³ serving
raw	none	24 000
traditional pasteurised	63°C for 30 minutes	
flash pasteurised	78°C for 35 seconds	1600
ultra-high temperature pasteurised (UHT)	135°C for 2 seconds	none

(i) Complete the table.

[2]

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(ii) Unopened UHT milk can be stored on supermarket shelves at room temperature for six months but unopened pasteurised milk must be stored in a refrigerator at 4°C or below for no longer than seven days. Explain these differences.

[4]

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